BOUNDED M_{RVM}: TRANSPORT OF ORDINATES AND IDENTIFICATION OF THE ENDPOINT

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Scope. This note reviews and corrects the three-step argument that transports the critical RVM product to the genuine ordinates. Steps 1 and 2 are correct and are stated with proofs; the identification theorem of Step 3 is correct as an equivalence; the verification proposed in Step 3 through Dirichlet coefficients of $1/\zeta_{\mathrm{RVM}}$ is not available, because those coefficients do not exist (Proposition 1.5). The notation and the transition theorem (TR) are those of the insert `TE_tracking_v57`.

Contents

1. The transported product and the identification of its endpoint 1
1.1. Step 1: the transported product 1
1.2. Step 2: the flow of ζ_{RVM} 2
1.3. Step 3: identification of the endpoint 2

1. The transported product and the identification of its endpoint

Let γ_k be the k-th genuine ordinate, counted with multiplicity and without assuming $\beta_k = \frac{1}{2}$; let $\tau_k = \gamma_k^{\mathrm{RVM}}$, defined by $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, and $\eta_k = \tau_k - \gamma_k$. Put $q(s) = s(s-1)$ and $G(s) = s(s-1)\Gamma(s/2)\pi^{-s/2}$.

1.1. Step 1: the transported product.

For $0 \le \lambda \le 1$ let $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$ and

$$P_\lambda(s) = \prod_{k\ge 1} \Big(1 + \frac{s(s-1)}{\frac{1}{4} + \gamma_k(\lambda)^2}\Big). \tag{1.1}$$

Theorem 1.1 (Transported product)**.** *The product* (1.1) *converges locally uniformly on* $[0,1]\times\mathbb{C}$*. Each* P_λ *is entire, depends continuously on* λ*, has zeros exactly at* $\frac{1}{2} \pm i\gamma_k(\lambda)$ *(with multiplicity), satisfies* $P_\lambda(1-s) = P_\lambda(s)$*,* $P_\lambda(\bar{s}) = \overline{P_\lambda(s)}$ *and* $P_\lambda(0) = P_\lambda(1) = 1$*, and* $P_\lambda^{(j)} \to P_0^{(j)}$ *locally uniformly as* $\lambda \to 0^+$ *for every* $j \ge 0$*. Moreover* $P_1 = P_{\mathrm{RVM}}^{\mathrm{crit}}$*.*

Proof. From $N(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(1/T)$ and $S(T) = O(\log T)$ one gets $\mathcal{N}_{\mathrm{RVM}}(\tau_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k) = O(\log\gamma_k)$; since $\mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, the mean value theorem gives $\eta_k = O(1)$, and $\gamma_k \asymp \tau_k \asymp k/\log k$. On a compact set K, $|q| \le C_K$ and $\frac{1}{4} + \gamma_k(\lambda)^2 \gg \gamma_k^2$ uniformly in λ for k large, so $\sup_{K\times[0,1]} |q/(\frac{1}{4}+\gamma_k(\lambda)^2)| \ll_K (\log k)^2 k^{-2}$, which is summable: the product converges normally. The zero statement follows from $\frac{1}{4} + \gamma_k(\lambda)^2 + s(s-1) = (s-\frac{1}{2})^2 + \gamma_k(\lambda)^2$, the symmetries from the dependence on $s(s-1)$ and the reality of $\gamma_k(\lambda)$, and $P_\lambda(0) = P_\lambda(1) = 1$ from $q(0) = q(1) = 0$. With $A_{k,\lambda} = \frac{1}{4} + \gamma_k(\lambda)^2$, $|q/A_{k,\lambda} - q/A_{k,0}| \ll_K \lambda|\eta_k|\gamma_k^{-3} + \lambda^2\eta_k^2\gamma_k^{-4}$, which is summable; hence $P_\lambda \to P_0$ locally uniformly, and Cauchy's formula gives the derivatives. □

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1.2. **Step 2: the flow of ζ_{RVM}.** Define

$$\zeta_\lambda = \frac{P_\lambda}{G}, \qquad \zeta_1 = \zeta_{\mathrm{RVM}}, \qquad \zeta_{Z1} := \zeta_0 = \frac{P_0}{G}, \qquad F_\lambda = \frac{1}{\zeta_\lambda}. \tag{1.2}$$

Theorem 1.2 (Flow)**.** *As $\lambda \to 0^+$, $\zeta_\lambda \to \zeta_{Z1}$ locally uniformly in the spherical metric, and $F_\lambda \to F_0$ uniformly on compact sets free of zeros of P_0. Off the moving divisor,*

$$\partial_\lambda \log P_\lambda(s) = \mathcal{V}_\lambda(s) := -2s(s-1) \sum_{k\geq 1} \frac{\gamma_k(\lambda)\eta_k}{(\frac{1}{4}+\gamma_k(\lambda)^2)(\frac{1}{4}+\gamma_k(\lambda)^2 + s(s-1))}, \tag{1.3}$$

the series converging locally uniformly, and

$$\zeta_{Z1} = \zeta_{\mathrm{RVM}} \frac{P_0}{P_1} = \zeta_{\mathrm{RVM}} \exp\Big(- \int_0^1 \mathcal{V}_u \, du\Big). \tag{1.4}$$

If a zero $\frac{1}{2}+i\gamma_k$ of P_0 is simple, the residues $G(\rho_{k,\lambda})/(\rho_{k,\lambda}P_\lambda'(\rho_{k,\lambda}))$ of $F_\lambda(s)/s$ at $\rho_{k,\lambda} = \frac{1}{2}+i\gamma_k(\lambda)$ converge to the corresponding residue of $F_0(s)/s$.

Proof. Uniform convergence of P_λ and division by the fixed function G give the first claim; on a compact set with $\min|P_0| = m > 0$ one has $|P_\lambda| \geq m/2$ for small λ and $|F_\lambda - F_0| \leq 2\sup|G|\, m^{-2}|P_\lambda - P_0|$. Termwise, $\partial_\lambda \operatorname{Log}(1+q/A_{k,\lambda}) = -2q\gamma_k(\lambda)\eta_k/(A_{k,\lambda}(A_{k,\lambda}+q)) = O_K(|\eta_k|\gamma_k^{-3})$, summable. For (1.4), integrating each mode with $u = \gamma_k(\lambda)$, $du = \eta_k\, d\lambda$, gives $\frac{1}{2q}\operatorname{Log}\frac{u^2+1/4}{u^2+1/4+q}\big|_{\gamma_k}^{\tau_k}$ modulo $2\pi i$ (the principal branch is valid only while the argument does not cross $(-\infty, 0]$); after exponentiation the factor $\prod_k (1+q/A_{k,0})/(1+q/A_{k,1}) = P_0/P_1$ is exact. The residues converge by Theorem 1.1. □

Identity (1.4) is the trivial factorization $P_0/G = (P_1/G)(P_0/P_1)$ written as a flow. The endpoint is determined directly by the genuine ordinates placed on the critical line:

$$\zeta_{Z1}(s) = \frac{1}{G(s)} \prod_{k\geq 1} \Big(1 + \frac{s(s-1)}{\frac{1}{4}+\gamma_k^2}\Big).$$

1.3. **Step 3: identification of the endpoint.** The Hadamard factorization of $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, pairing ρ with $1-\rho$ and using $2\xi(0) = 1$, gives

$$2\xi(s) = \prod_{\rho:\ \Im\rho > 0} \Big(1 + \frac{s(s-1)}{\rho(1-\rho)}\Big), \tag{1.5}$$

the product taken in order of increasing $\Im\rho$.

Proposition 1.3 (Normalization of the endpoint)**.** *$P_0 = 2\xi$ if and only if every nontrivial zero has real part $\frac{1}{2}$. Equivalently, $\zeta_{Z1} = \zeta$ if and only if RH holds.*

Proof. For $\rho = \beta + i\gamma$, $\rho(1-\rho) = \beta(1-\beta) + \gamma^2 + i\gamma(1-2\beta)$, which equals $\frac{1}{4}+\gamma^2$ if and only if $\beta = \frac{1}{2}$. If RH holds, the factors of (1.5) and of P_0 coincide and $P_0 = 2\xi$. Conversely, if $P_0 = 2\xi$ the two functions have the same zeros, and the zeros of P_0 lie on $\Re s = \frac{1}{2}$. Since $\zeta = 2\xi/G$ and $\zeta_{Z1} = P_0/G$, the second statement follows. The constant is fixed by $P_0(0) = 1 = 2\xi(0)$. □

For $c > 1$ put $M_{Z1}(x) = \lim_{U\to\infty} \frac{1}{2\pi i}\int_{c-iU}^{c+iU} F_0(s)\frac{x^s}{s}\, ds$ whenever the limit exists.

Theorem 1.4 (Identification)**.** *The following are equivalent:*

(i) *M_{Z1} exists and $M_{Z1}(n) = M(n)$ for every $n \geq 1$;*
(ii) *$1/\zeta_{Z1}$ is, in $\Re s > 1$, the Dirichlet series $\sum_n \mu(n)n^{-s}$;*
(iii) *$\zeta_{Z1} = \zeta$;*
(iv) *RH.*

Proof. (i)⇒(ii): the normalized Mellin transform $s\int_1^\infty M(x)x^{-s-1}dx = 1/\zeta(s)$ coincides with $1/\zeta_{Z1}$ for $\Re s > c$. (ii)⇒(iii): identity of meromorphic functions. (iii)⇔(iv): Proposition 1.3. (iii)⇒(i): Perron's formula for $1/\zeta$. □

Proposition 1.5 (No Dirichlet expansion for ζ_{RVM})**.** *Neither ζ_{RVM} nor $1/\zeta_{\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in any half-plane $\Re s > \sigma_0$.*

Proof. In the proof of the transition theorem (TR) of `TE_tracking_v57`, the restriction $\sigma' < \frac{1}{2}$ ($s = \frac{1}{2}+\sigma'+it$) is used only in the passage from w to χ; the form $\zeta_{\mathrm{RVM}}(s) = Z_{\mathrm{RVM}}(1+w)(1+O(1/t))$, with $w = e^{-2\pi i\mathcal{N}_{\mathrm{RVM}}(t-i\sigma')}$, holds uniformly on every compact range of $\sigma' > 0$ (in the bound on the rays it suffices that $t^2 \geq 2(\tau_0^2 + \sigma'^2)$). For fixed $\sigma' > 0$, $\Im\mathcal{N}_{\mathrm{RVM}}(t - i\sigma') = -\frac{\sigma'}{2\pi}\log\frac{t}{2\pi} + O(t^{-2}) \to -\infty$, so $w \to 0$ and $1/\zeta_{\mathrm{RVM}}(\sigma + it) \to L = 1/Z_{\mathrm{RVM}}$ as $|t| \to \infty$, for every fixed $\sigma > \frac{1}{2}$. If $1/\zeta_{\mathrm{RVM}} - L = \sum_n c(n)n^{-s}$ converged absolutely on $\Re s = \sigma > \sigma_0$, the mean value theorem for absolutely convergent Dirichlet series would give

$$\sum_{n\geq 1}|c(n)|^2 n^{-2\sigma} = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\Big|\frac{1}{\zeta_{\mathrm{RVM}}(\sigma+it)} - L\Big|^2 dt = 0,$$

so $1/\zeta_{\mathrm{RVM}} \equiv L$, contradicting $1/\zeta_{\mathrm{RVM}}(1) = 0$. The same argument with $\zeta_{\mathrm{RVM}} \to Z_{\mathrm{RVM}}$ excludes ζ_{RVM}. □

Consequently the coefficients $a_{\mathrm{RVM}}(n)$ with $1/\zeta_{\mathrm{RVM}} = \sum a_{\mathrm{RVM}}(n)n^{-s}$ do not exist, and under RH neither do coefficients $\kappa(n)$ of $\exp\int_0^1 \mathcal{V} = \zeta_{\mathrm{RVM}}/\zeta$, since otherwise $\zeta_{\mathrm{RVM}} = \zeta\cdot\sum\kappa(n)n^{-s}$ would be a Dirichlet series. The proposed check $(a_{\mathrm{RVM}} * \kappa)(n) = \mu(n)$ is therefore not a verifiable condition.

Remark 1.6 (Status). Proved: the transport $\zeta_{\mathrm{RVM}} = \zeta_1 \to \zeta_\lambda \to \zeta_{Z1} = P_0/G$, locally uniformly and with divisors and residues transported, and the equivalence $M_{Z1} = M \iff \zeta_{Z1} = \zeta \iff P_0 = 2\xi \iff$ RH. Not proved: $M_{Z1} = M$. The endpoint depends only on the ordinates γ_k, while the content of RH lies in the abscissae β_k, which the transport does not move: $\zeta_{Z1} = \zeta$ exactly when every factor $1 + s(s-1)/(\rho(1-\rho))$ of (1.5) equals the corresponding factor $1 + s(s-1)/(\frac{1}{4}+\gamma^2)$ of P_0.